

THEMIS

An Integrated Digital Platform for Legal Aid, Document Automation,
and Case Management in India

Project Report

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Academic Year: 2025–2026

Abstract

Access to justice within the Indian legal ecosystem remains a significant challenge due to increasing case backlogs, procedural complexity, and limited access to reliable legal information. Citizens seeking legal recourse are frequently unable to navigate the system on their own, while legal aid organizations and advocates lack a unified digital infrastructure to manage cases, verify eligibility, and reach those who need assistance most. **Themis** is a comprehensive full-stack legal technology platform developed to simplify legal processes, improve digital legal literacy, and enhance access to legal aid services for citizens across this ecosystem.

The platform provides an integrated environment for legal knowledge discovery, guided legal assessments, automated document generation, secure document management, and case tracking. A citizen can search legal information relevant to their situation, work through a guided assessment that clarifies which legal remedies apply to them, and generate editable legal complaints or Right to Information (RTI) applications without needing to draft them from scratch or consult an advocate for routine paperwork. Supporting documents can be uploaded and securely stored against a case, and the progress of legal proceedings can be monitored through an interactive timeline interface that gives citizens visibility into a process that is otherwise opaque and difficult to track. The system also incorporates a verification mechanism that connects eligible citizens with verified legal aid advocates, thereby improving transparency, accessibility, and efficiency in legal service delivery, rather than leaving that connection to informal referrals.

The backend is built using a modern TypeScript technology stack consisting of **Node.js**, **Hono**, **tRPC**, **Drizzle ORM**, and **PostgreSQL**, ensuring high performance, scalability, and end-to-end type safety between the client and server. Authentication is implemented using **JSON Web Tokens (JWT)**, while **Zod** is utilized for robust schema validation at every API boundary, reducing the class of runtime errors that typically arise in looser, untyped systems. The frontend is developed using **React**, **TypeScript**, **Vite**, **Tailwind CSS**, **shadcn/ui**, **Radix UI**, and **Lucide React**, delivering a responsive, accessible, and user-friendly interface across desktop and mobile devices, which matters particularly for a platform meant to serve citizens with varying levels of digital literacy.

Themis integrates intelligent document automation, secure data management, and modern web technologies into a unified platform capable of supporting legal aid organizations, advocates, and citizens alike. The proposed system demonstrates a scalable and production-ready solution for modernizing public legal aid services in India while promoting efficient collaboration, secure information exchange, and improved access to justice for the populations that need it most.

Technologies Used

Frontend Technologies

Technology	Purpose
React	Component-based user interface development
TypeScript	Type-safe JavaScript development
Vite	Fast build tool and development server
Tailwind CSS	Utility-first CSS framework
shadcn/ui	Modern reusable UI components
Radix UI	Accessible headless UI library
Lucide React	Icon library
React Context & Hooks	State management
npm	Package management
ESLint	Code quality and linting
Prettier	Code formatting
Git & GitHub	Version control and collaboration

Backend Technologies

Technology	Purpose
Node.js	JavaScript runtime environment
TypeScript	Backend development
Hono	Lightweight web framework
tRPC	End-to-end type-safe API communication
Drizzle ORM	Object relational mapping (ORM)
PostgreSQL	Relational database management system
Zod	Schema validation
JWT	Authentication and authorization

Key Features

- Advanced legal information search
- Guided legal assessments
- Automated complaint generation
- Automated Right to Information (RTI) application generation
- Secure document upload and management
- Case and hearing timeline tracking
- Advocate verification and legal aid matching
- Responsive and accessible user interface
- End-to-end type-safe full-stack architecture